Software Requirement Specification

NeuroRead

Inclusive Digital Library · Agentic AI Reading Assistant

Introduction

Purpose

NeuroRead is an AI-powered inclusive digital reading platform designed to support neurodivergent learners — including those with Dyslexia, ADHD, and Visual Impairment — while improving reading comprehension for all users. Traditional PDF readers and digital libraries are passive, one-size-fits-all tools that do not account for diverse cognitive needs. Students with learning differences struggle with dense academic PDFs, tight typography, lack of focus assistance, and zero proactive guidance, resulting in cognitive overload, reduced engagement, and poor academic performance.

NeuroRead addresses this gap by combining an intelligent Agentic AI engine with accessibility-first design. The platform dynamically adapts typography, simplifies complex text, provides focus-enhancing tools, and proactively assists users before they even ask — making digital reading intelligent, inclusive, and adaptive.

The main purpose of this document is to provide a complete and detailed description of the software requirements for NeuroRead, intended for use by developers, testers, and project evaluators. Any future changes to requirements will follow a formal change approval process.

Scope

This SRS document describes the complete functional and non-functional requirements of NeuroRead, covering the following scope:

- A browser-based reading platform accessible on desktop and mobile via standard web browsers.
- Built-in curated digital library with 12+ books across fiction, science, history, self-help, and more.
- User file upload and management for PDF, TXT, Markdown, HTML, and RTF formats.
- Camera document capture with OCR-based text extraction.
- Personalised Reading DNA profiling with a 5-question adaptive quiz.
- Full typography customisation: font family, size, letter spacing, line height, word spacing, and themes.
- AI-powered adaptive suggestions based on scroll behaviour, pauses, and back-scroll detection.
- Voice command navigation for hands-free control.

• Text-to-Speech (TTS) playback with word-level highlighting and speed control.

• Reading progress tracking, highlight storage, bookmarking, notes, and session history.

• Multilingual interface switching (English, Hindi, Kannada) for regional accessibility.

Boundaries: The current scope does not include native mobile apps, real-time eye tracking, or full LMS integration. These are identified as future enhancements.

Target Users: Students with Dyslexia, ADHD learners, visually impaired readers, competitive exam aspirants, and educational institutions.

Overall Description

NeuroRead is an implementation of an intelligent inclusive reading platform. The system holds a curated digital library, supports user-uploaded documents, and applies personalised reading profiles. The Admin role manages the library and user data, while the Reader role benefits from personalised accessibility settings.

The database connectivity is planned using Django REST Framework with SQLite3. Security and data protection are implemented via JWT-based token authentication.

NeuroRead uniquely adds Agentic AI that proactively detects reading difficulty and offers real-time personalised interventions, along with India-specific regional language support.

Problem Statement

Students and readers in India and globally face the following challenges:

- Dense academic PDFs and long-form content cause cognitive overload.
- Dyslexia (affecting ~10–15% of the population) significantly reduces reading fluency and comprehension.
- ADHD causes loss of focus and difficulty sustaining attention in long-text environments.
- Visually impaired users require high-contrast, scalable interfaces that most readers do not provide.
- Traditional PDF readers are entirely passive — they provide no intelligence, suggestions, or adaptation.
- No mainstream platform offers proactive AI-powered reading assistance or cognitive profiling.

Result: Millions of learners experience reduced academic efficiency not due to lack of intelligence, but because their tools are not designed for the diversity of human cognition.

Product Functions

The website will allow access only to authorised users with specific roles:

- **Administrator:** Maintains the platform, manages the library, and views user details.
- **Reader (User):** Registers a profile, uploads documents, sets Reading DNA, reads with personalised settings.

Reader Role

On the registration form, the reader enters their name, email, password, and accessibility preferences. The system assigns a Reading DNA profile based on a 5-question quiz and applies adaptive typography settings automatically.

Administration Role

The system administrator can: add, update, and remove books from the library; manage user accounts; view reading activity reports; and configure global accessibility settings.

Special Requirements

Interfaces

- Admin Interface: Manage library, users, and system configuration.
- User Interface: Personalised, accessibility-first reading environment.

The application will have a user-friendly, menu-based, and keyboard-navigable interface.

Hardware Requirements

Component	Minimum Specification
Processor	Intel Core i3 (or equivalent) — 1.5 GHz or higher
RAM	4 GB minimum (8 GB recommended)
Hard Disk	500 MB free space for application and database
Display	1024 × 768 resolution minimum (1920 × 1080 recommended)
Camera (optional)	Any webcam or device camera for document capture feature

Software Requirements

Component	Specification
Operating System	Windows 10/11, macOS 12+, or Ubuntu 20.04+
Browser	Google Chrome 110+, Mozilla Firefox 110+, Safari 16+
Python	Python 3.10+ (for Django backend)
Node.js (optional)	v18+ for front-end tooling
Database	SQLite3 (bundled with Python)

Server Side

A Django-based web server will accept all requests from the client. A development database will be hosted locally (SQLite3); the production database is hosted centrally. JWT authentication secures all API endpoints.

Tools & Technologies Used

Frontend Technologies

Technology	Purpose
HTML5	Structure and semantic markup
CSS3 (Custom Properties)	Dynamic theming, adaptive typography, responsive layout
Vanilla JavaScript (ES6+)	All interactive logic, state management, adaptive engine
PDF.js (v3.11)	Client-side PDF text extraction and page rendering

Backend Technologies

Technology	Purpose
Django (Python)	Server-side framework, REST API backend
Django REST Framework	API endpoint construction and serialization
SQLite3	Local relational database for user data, progress, and files
JWT Authentication	Secure token-based user authentication

AI Layer

Technology	Purpose
HuggingFace Inference API	Cloud AI inference for text complexity analysis and summarisation
distilbart model	Abstractive text simplification for neurodivergent readers
deep-translator	Multilingual translation (English, Hindi, Kannada, and more)
Tesseract.js	OCR for camera-captured physical document text extraction

Development Tools

Tool	Purpose
VS Code	Primary IDE for development
Git & GitHub	Version control and collaboration
Chrome DevTools	Debugging, accessibility testing, performance profiling

Functional Requirements

The functional requirements of NeuroRead are divided among the Reader (User) and Administrator roles of the application.

Use Case: User Registration

Field	Detail
Description	This use case describes the scenario where the user creates an account to access the platform.
Actor	User / Reader
Input	Personal details: Name, Email, Password, Accessibility preferences
Output	All details are saved to the database. A Reading DNA quiz is triggered to set the initial profile.

Use Case: Reading DNA Profiling

Field	Detail
Description	A 5-question adaptive quiz assesses the user's cognitive reading profile across letter clarity, line tracking, background preference, spacing sensitivity, and reading pace.
Actor	User / Reader
Input	Answers to 5 adaptive quiz questions
Output	System assigns one of four profiles and automatically applies optimised typography settings.

Use Case: Browse & Read Library Books

Field	Detail
Description	The user browses the built-in library, filters by category or search query, and opens a book to read.
Actor	User / Reader
Input	Category filter or search keyword
Output	Displays the selected book with personalised typography and reading tools active.